

Test Report

Applicant : Makerfabs
Address : 6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Product Name : UWB Radio Frequency Module
Brand Mark : 
Model : MaUWB-DW3000 CE
Series model : N/A
Report Number : BLA-EMC-202607-A39-04
Date of Receipt : Jul. 20, 2026
Date of Test : Jul. 20, 2026 to Aug. 04, 2026
Test Standard : ETSI EN 301 489-1 V2.2.3 (2019-11)
ETSI EN 301 489-33 V2.2.1(2019-04)
Test Result : Pass

Compiled by: Mark Chen

Review by: Xavier

Approved by: 

Issued Date: Aug. 31, 2026

BlueAsia Technical Services (Shenzhen) Co., Ltd

Address: No.41, South of Beihuan Road, Shangwu Community, Shiyuan Subdistrict, Bao'an District, Shenzhen, Guangdong, China



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Revise Record

Version No.	Date	Description
01	Aug. 31, 2026	Original

1 General information

1.1 General information

Applicant	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Manufacturer	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Factory	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China

1.2 General description of EUT

Product Name	UWB Radio Frequency Module
Model No.	MaUWB-DW3000 CE
Operation Frequency	6.0GHz to 8.5GHz
Power supply	DC 3.3V
Hardware version	N/A
Software version	N/A
<i>Note: For a more detailed description, please refer to Specification or User's Manual supplied by the applicant and/or manufacturer.</i>	

2 Test summary

No.	Test item	Test Requirement	Test Method	Class/Severity	Result
1	Radiated Emissions (30MHz-1GHz)	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)	CENELEC EN 55032 (2015)	Class B	Pass
2	Radiated Emissions (Above 1GHz)	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)	CENELEC EN 55032 (2015)	Class B	Pass
3	Electrostatic Discharge	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)	CENELEC EN 61000-4-2 (2009)	4kV Contact Discharge 8kV Air Discharge	Pass
4	Radiated Immunity	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)	CENELEC EN 61000-4-3 (2006), A1 (2008) and A2 (2010)	3V/m, 80%, 1kHz Amp. Mod.	Pass

N/A: Not Applicable

3 Test Configuration

3.1 Test mode

Test mode	Test mode description
TM1	Keep the EUT in UWB mode
Remark: only the data of the worst mode would be recorded in this report.	

3.2 Test environment

Ambient Temperature:	15 °C to 35 °C
Relative Humidity:	25 % to 75 %
Atmospheric pressure:	86 kPa to 106 kPa

3.3 Auxiliary equipment

Device Type	Manufacturer	Model Name	Serial No.	Remark
Rechargeable battery	TIANNENG	6-DZF-20.3	N/A	From lab
Note: "--" mean no any auxiliary device during testing.				

4 Laboratory information

4.1 Laboratory and accreditations

The test facility is recognized, certified, or accredited by the following organizations:

Company name:	BlueAsia Technical Services (Shenzhen) Co.,Ltd.
Address:	No.41, South of Beihuan Road, Shangwu Community, Shiyao Subdistrict, Bao'an District, Shenzhen, Guangdong, China
CNAS accredited no.:	L9788
Telephone:	+86-755-28682673
FAX:	+86-755-28682673

4.2 Measurement uncertainty

Parameter	Expanded Uncertainty (Confidence of 95%)
Radiated Emission(9kHz-30MHz)	$\pm 4.34\text{dB}$
Radiated Emission(30MHz-1000MHz)	$\pm 4.24\text{dB}$
Radiated Emission(1GHz-18GHz)	$\pm 4.68\text{dB}$
AC Power Line Conducted Emission(150kHz-30MHz)	$\pm 3.45\text{dB}$

5 Test equipment

Radiated Spurious Emissions (Below 1GHz)

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-002-01	Anechoic chamber	9*6*6 chamber	SKET	N/A	2024/3/27	2027/3/26
BLA-EMC-002-02	Control room	966 control room	SKET	N/A	2024/3/27	2027/3/26
BLA-EMC-009	EMI receiver	ESR7	R&S	101199	2025/08/05	2026/08/04
BLA-EMC-043	Loop antenna	FMZB1519B	Schwarzbeck	00102	2026/05/31	2027/05/30
BLA-EMC-065	Broadband antenna	VULB9168	Schwarzbeck	01065P	2026/05/31	2027/05/30
BLA-XC-01	Coaxial Cable	N/A	BlueAsia	V01	N/A	N/A
BLA-XC-02	Coaxial Cable	N/A	BlueAsia	V02	N/A	N/A

Radiated Spurious Emissions (Above 1GHz)

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-001-01	Anechoic chamber	9*6*6 chamber	SKET	N/A	2023/11/16	2026/11/15
BLA-EMC-001-02	Control Room	966 control room	SKET	N/A	2025/11/12	2026/11/11
BLA-EMC-008	Spectrum	FSP40	R&S	100817	2025/08/05	2026/08/04
BLA-EMC-013	Horn Antenna	BBHA9120D	Schwarzbeck	01892	2026/05/31	2027/05/30
BLA-EMC-014	Amplifier	PA_000318G-45	SKET	PA2018043003	2025/08/05	2026/08/04
BLA-EMC-046	Filter bank	2.4G/5G Filter bank	SKET	N/A	2026/05/25	2027/05/24
BLA-EMC-061	Receiver	ESPI7	R&S	101477	2026/05/25	2027/05/24
BLA-EMC-066	Amplifier	LNPA_30M01G-30	SKET	SK2021060801	2026/05/25	2027/05/24
BLA-EMC-086	Amplifier	LNPA_18G40G-50dB	SKET	SK2022071301	2026/05/25	2027/05/24
BLA-EMC-087	Horn Antenna	BBHA 9170	Schwarzbeck	1106	2026/05/31	2027/05/30
BLA-XC-03	Coaxial Cable	N/A	BlueAsia	V03	N/A	N/A
BLA-XC-04	Coaxial Cable	N/A	BlueAsia	V04	N/A	N/A

Radiated Immunity

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-058	Power Amplifier	HAP_80M01G-250W	SKET	N/A	2026/05/25	2027/05/24
BLA-EMC-059	Power Amplifier	HAP_01G03G-75W	SKET	N/A	2026/05/25	2027/05/24
BLA-EMC-060	Power Amplifier	HAP_03G06G-50W	SKET	N/A	2026/05/25	2027/05/24
BLA-EMC-067	Stacked log periodic antenna	STLP 9129 Plus	SKET	N/A	N/A	N/A
BLA-EMC-069	Field Probe	EP-601	Narda	811ZX10335	2025/08/05	2026/08/04
BLA-EMC-070	Power sensor	NRP-Z11	R&S	1138.3004.02-	2025/08/05	2026/08/04

				116101-Dr		
BLA-EMC-071	Power sensor	NRP-Z11	R&S	1138.3004.02-116946-KH	2025/08/05	2026/08/04
BLA-EMC-072	Signal Generator	N5181A	Aglient	MY46240904	2026/05/25	2027/05/24
BLA-EMC-088	Audio Analyzer	ATS-1	Audio Precision	ATS141094	2026/05/25	2027/05/24
BLA-EMC-090	Audio shielding box	SB-ABT-C35	SKET	N/A	N/A	N/A
BLA-EMC-091	1kHz calibration audio source	MCS-ABT-C35	SKET	N/A	N/A	N/A
BLA-EMC-092	Free Field Microphone	MGS MP 663	SKET	0414	N/A	N/A
BLA-XC-04	Coaxial Cable	N/A	BlueAsia	N/A	N/A	N/A

Electrostatic Discharge

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-024	Electrostatic discharge generator	ESS-2002	Noiseken	ESS03X2235	2025/08/11	2026/08/10

Test Software Record:

Software No.	Software Name	Manufacture	Software version	Test site
BLA-EMC-S001	EZ-EMC	EZ	EEMC-3A1+	RE
BLA-EMC-S002	EZ-EMC	EZ	EEMC-3A1+	RE
BLA-EMC-S009	EMC-S	SKET	V2.1.2.19	RS

6 Test result

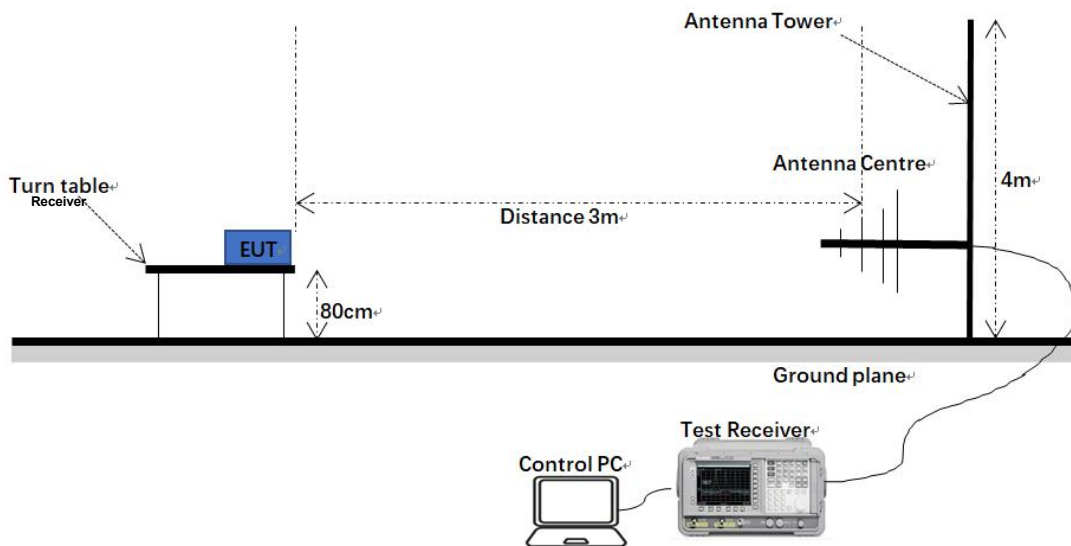
6.1 Radiated Emissions (30MHz-1GHz)

Test Standard	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)
Test Method	CENELEC EN 55032 (2015)
Test Mode (Pre-Scan)	TM1
Test Mode (Final Test)	TM1

6.1.1 Limits

Frequency Range	Limit
30MHz-230MHz	40 dB(μ V/m) quasi-peak
230MHz-1GHz	47 dB(μ V/m) quasi-peak

6.1.2 Test setup



Description of test setup connection:

- EUT was put on turn table in a 966 chamber room;
- The center distance of the receiving antenna is 3 meters from the EUT;
- The signal received by the receiving antenna is transmitted to the receiver through the coaxial line;
- Test receiver connected to control PC.

6.1.3 Procedure

An initial pre-scan was performed in the chamber using the spectrum analyzer in peak detection mode. Quasi-peak measurements were conducted based on the peak sweep graph. The EUT was measured by BiConiLog antenna with 2 orthogonal polarities.

Remark:

1. Final Level = Receiver Read level + Correct factor
2. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
3. The emission levels of other frequencies are very lower than the limit and not show in test report.

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6.1.4 Test data

[Test Mode: TM1]; [Polarity: Horizontal]



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector
1	72.5916	41.50	-17.07	24.43	40.00	-15.57	QP
2 *	87.1117	47.59	-20.62	26.97	40.00	-13.03	QP
3	130.8369	40.51	-17.23	23.28	40.00	-16.72	QP
4	161.4742	40.40	-15.54	24.86	40.00	-15.14	QP
5	178.1327	40.41	-16.96	23.45	40.00	-16.55	QP
6	779.6068	30.61	-5.56	25.05	47.00	-21.95	QP

Test Result: Pass

[Test Mode: TM1]; [Polarity: Vertical]



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector
1	40.1347	50.92	-16.82	34.10	40.00	-5.90	QP
2	75.7114	48.96	-18.04	30.92	40.00	-9.08	QP
3 *	87.7248	56.19	-20.74	35.45	40.00	-4.55	QP
4	131.7577	45.59	-17.13	28.46	40.00	-11.54	QP
5	492.4685	31.00	-11.31	19.69	47.00	-27.31	QP
6	776.8778	31.21	-5.62	25.59	47.00	-21.41	QP

Test Result: Pass

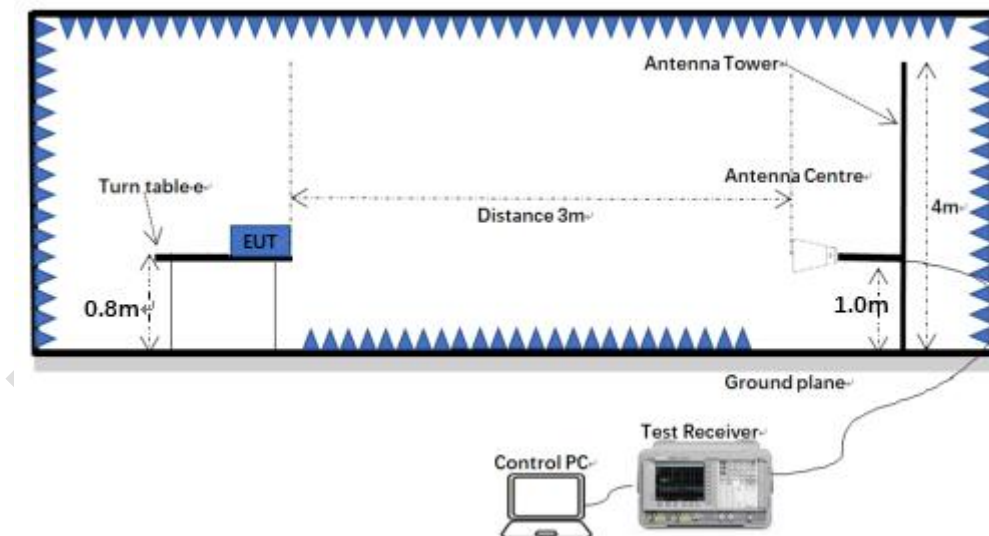
6.2 Radiated Emissions (above 1GHz)

Test Standard	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)
Test Method	CENELEC EN 55032 (2015)
Test Mode (Pre-Scan)	TM1
Test Mode (Final Test)	TM1

6.2.1 Limits

Frequency Range	Limit
1GHz-3GHz	70 dB(μ V/m) peak, 50dB(μ V/m) average
3GHz-6GHz	74 dB(μ V/m) peak, 54dB(μ V/m) average

6.2.2 Test setup



Description of test setup connection:

- EUT was put on turn table in a 966 chamber room;
- The center distance of the receiving antenna is 3 meters from the EUT;
- The signal received by the receiving antenna is transmitted to the receiver through the coaxial line;
- Test receiver connected to control PC.

6.2.3 Procedure

An initial pre-scan was performed in the chamber using the spectrum analyser in peak detection mode. Average measurements were conducted based on the peak sweep graph. The EUT was measured by Horn antenna with 2 orthogonal polarities.

Remark:

1. Final Level = Receiver Read level + Correct factor
2. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
3. The emission levels of other frequencies are very lower than the limit and not show in test report.

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6.2.4 Test data

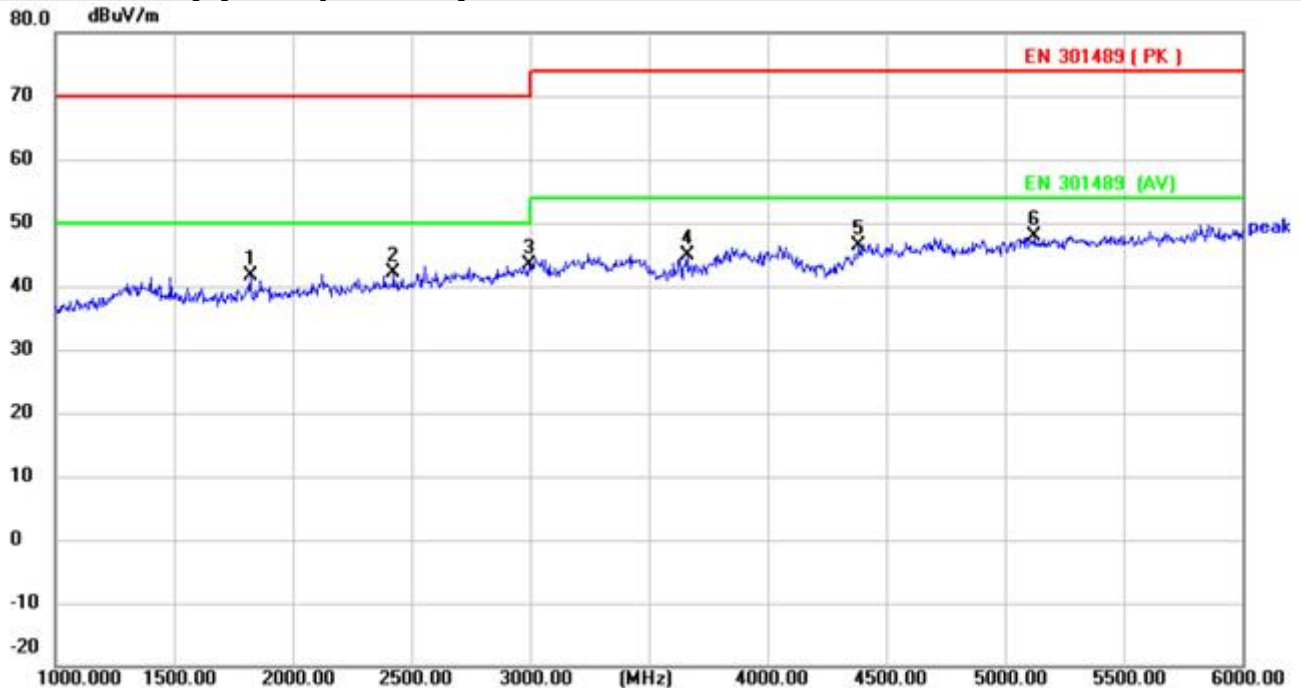
[Test Mode: TM1]; [Polarity: Horizontal]



No.	Mk.	Freq.	Reading Level	Correct Factor	Measurement	Limit	Over	
		MHz	dBuV	dB	dBuV/m	dBuV/m	dB	Detector
1		1500.000	53.81	-13.73	40.08	70.00	-29.92	peak
2		2120.000	53.67	-11.72	41.95	70.00	-28.05	peak
3		2520.000	52.40	-9.71	42.69	70.00	-27.31	peak
4		3245.000	51.50	-6.39	45.11	74.00	-28.89	peak
5		3830.000	49.71	-4.22	45.49	74.00	-28.51	peak
6	*	4730.000	48.09	-0.56	47.53	74.00	-26.47	peak

Test Result: Pass

[Test Mode: TM1]; [Polarity: Vertical]



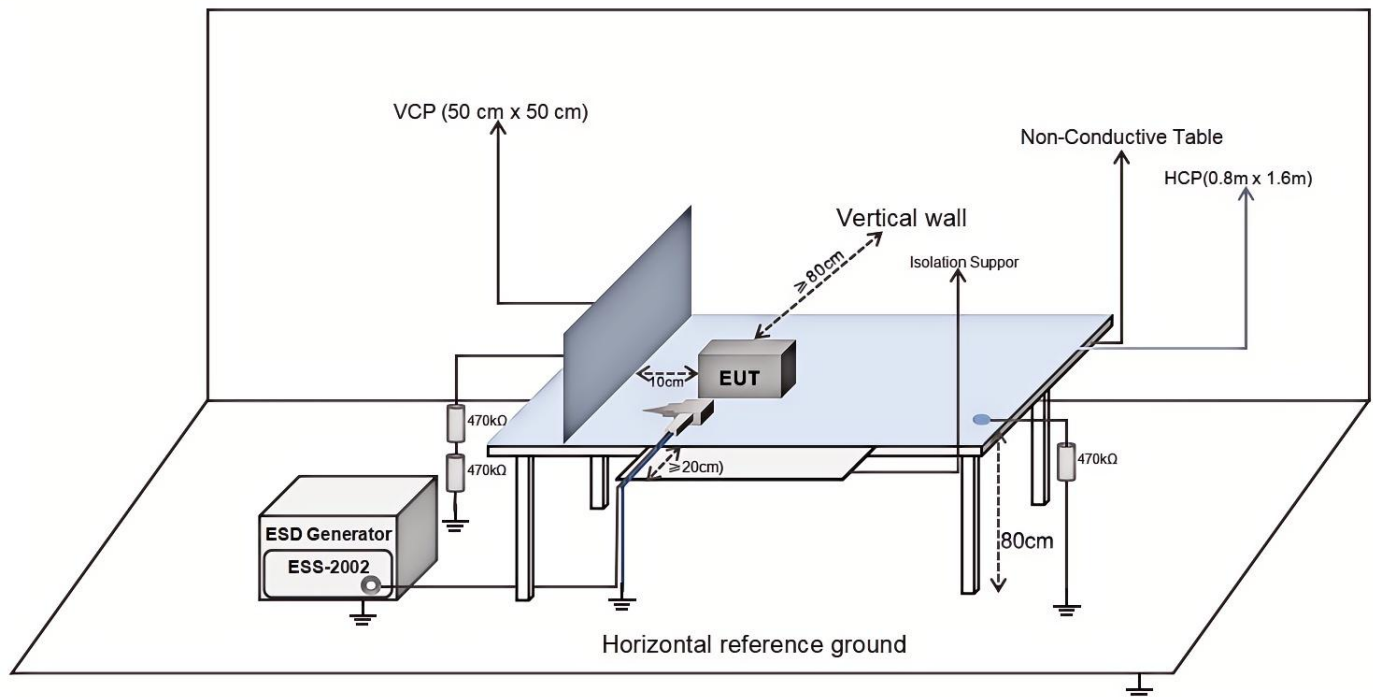
No.	Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB	Measure- ment dBuV/m	Limit dBuV/m	Over dB	Detector
1		1820.000	54.74	-13.06	41.68	70.00	-28.32	peak
2		2425.000	52.45	-10.25	42.20	70.00	-27.80	peak
3		2995.000	50.39	-7.06	43.33	70.00	-26.67	peak
4		3660.000	49.71	-4.94	44.77	74.00	-29.23	peak
5		4385.000	48.36	-2.08	46.28	74.00	-27.72	peak
6	*	5125.000	47.49	0.31	47.80	74.00	-26.20	peak

Test Result: Pass

6.3 Electrostatic Discharge

Test Standard	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)
Test Method	CENELEC EN 61000-4-2 (2009)
Test Level	4kV Contact Discharge 8kV Air Discharge
Performance Criterion:	B
Discharge Impedance:	330Ω/150pF
Number of Discharge:	Minimum 10 times at each test point
Discharge Mode:	Single Discharge
Discharge Period:	1 second minimum
Test Mode (Pre-Scan)	TM1
Test Mode (Final Test)	TM1

6.3.1 Test setup



6.3.2 Performance

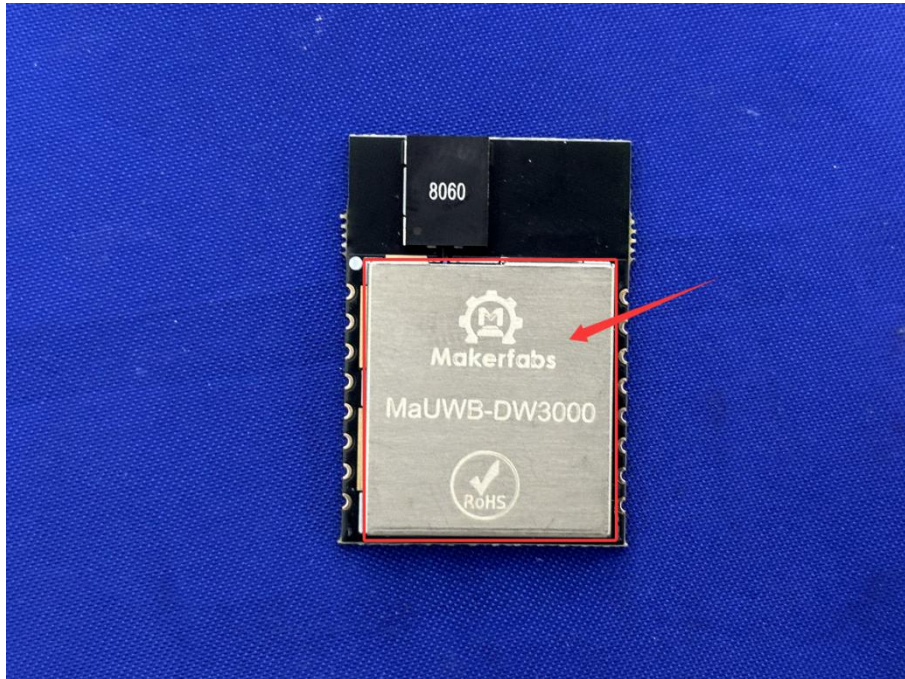
Criterion A	The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion B	After the test, the equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test. If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion C	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.
Criterion D	Loss of function or degradation of performance which is not recoverable, owing to damage to hardware or software, or loss of data.

6.3.3 Test data

Test mode	Level	Test points	Observations (Performance Criterion)	Result
Air discharge	±2kV	--	--	--
Air discharge	±4kV	--	--	--
Air discharge	±8kV	--	--	--
Contact discharge	±2kV	Metal	A	Pass
Contact discharge	±4kV	Metal	A	Pass
indirect discharge	±2kV	HCP	A	Pass
indirect discharge	±4kV	HCP	A	Pass
indirect discharge	±2kV	VCP	A	Pass
indirect discharge	±4kV	VCP	A	Pass

6.3.4 Test point

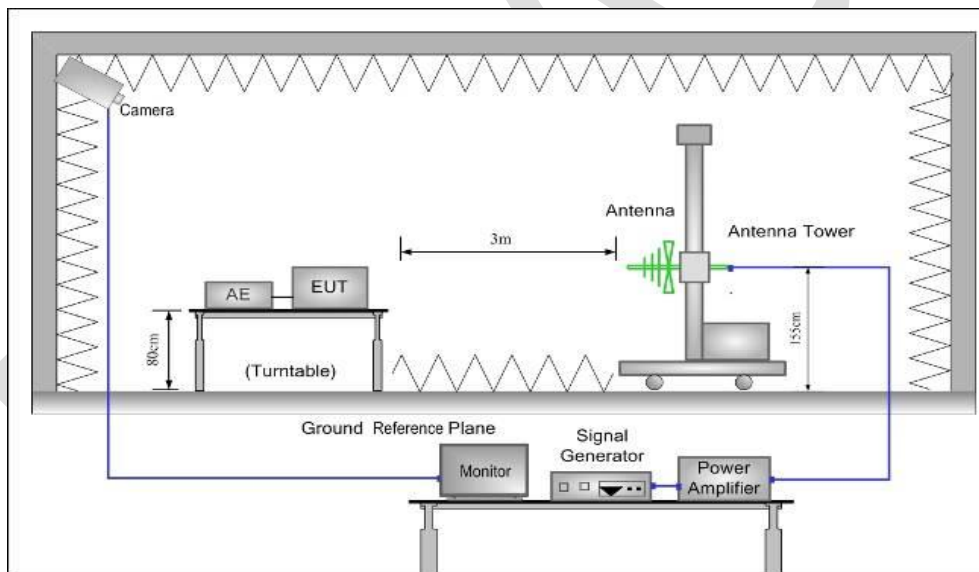
Contact discharge



6.4 Radiated Immunity

Test Standard	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-33 V2.2.1(2019-04)
Test Method	CENELEC EN 61000-4-3 (2006), A1 (2008) and A2 (2010)
Test Level	3V/m, 80%, 1kHz Amp. Mod.
Performance Criterion:	A
Frequency Range:	80MHz to 6GHz
Antenna Polarization:	Vertical and Horizontal
Modulation	1kHz, 80% Amp. Mod, 1% increment
Test Mode (Pre-Scan)	TM1
Test Mode (Final Test)	TM1

6.4.1 Test setup



6.4.2 Performance

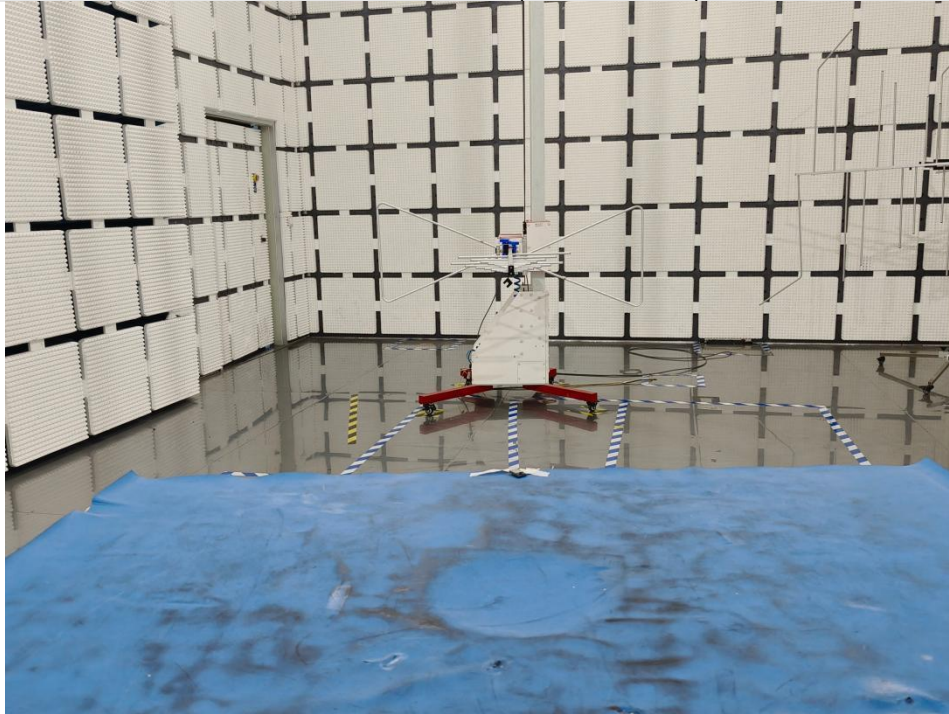
Criterion A	The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion B	After the test, the equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test. If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion C	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.
Criterion D	Loss of function or degradation of performance which is not recoverable, owing to damage to hardware or software, or loss of data.

6.4.3 Test data

Frequency	Level	Modulation	Antenna Polarization	EUT Face	Observations (Performance Criterion)	Result
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Front	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Front	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Rear	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Rear	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Left	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Left	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Right	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Right	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Top	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Top	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	V	Bottom	A	Pass
80MHz-6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment, dwell time=3seconds	H	Bottom	A	Pass

Appendix A: photographs of test setup

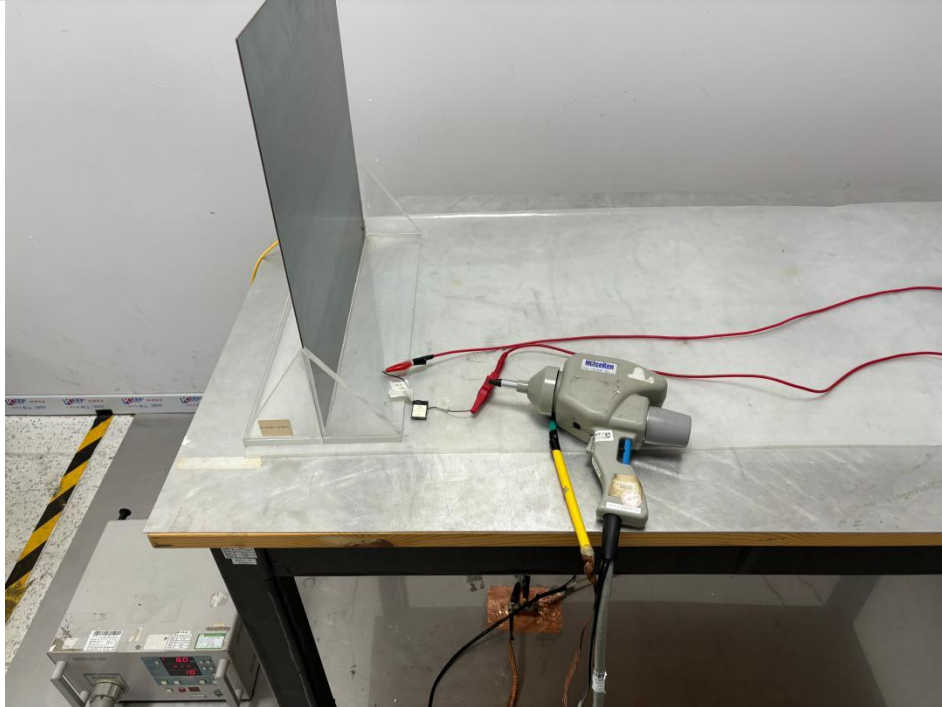
Radiated Emissions (30MHz-1GHz)



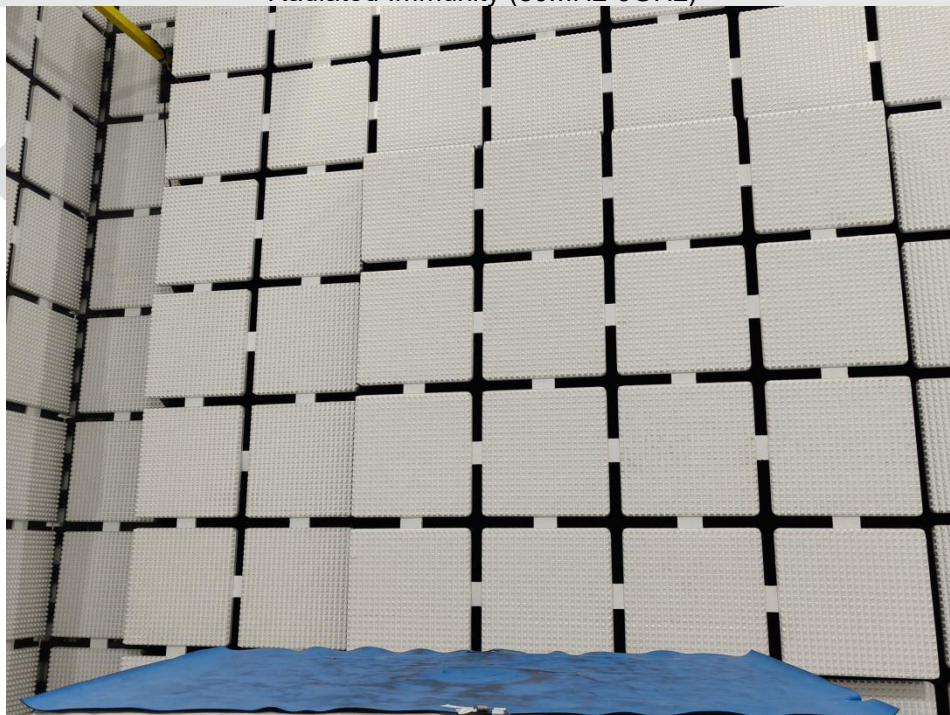
Radiated Emissions (Above 1GHz)



Electrostatic Discharge



Radiated Immunity (80MHz-6GHz)



Appendix B: photographs of EUT

Reference to the test report no. BLA-EMC-202607-A39-01.

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----END OF REPORT----

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